

Tanager VSWIR reveals mine-waste and alteration minerals that broadband satellites blur

Planet Tanager Open Data Competition entry. Alex Bradley, Department of Earth, Environmental, and Planetary Sciences, Washington University in St. Louis (abradley@wustl.edu).

The question

What can Tanager's 426-band visible-to-shortwave-infrared (VSWIR) spectra tell us about mine waste that broadband satellite imagery cannot? Hydrothermal alteration and mine waste are recorded in a small set of minerals — alunite, kaolinite, jarosite, the iron oxides, gypsum, and the white micas — whose diagnostic absorptions fall in the shortwave infrared. We test whether Tanager preserves those signatures where Sentinel-2 blurs them, and whether they can help identify mine waste showing evidence of acidic oxidative weathering, at two well-studied districts: the Bingham Canyon / Kennecott porphyry-copper mine and tailings impoundment in Utah, and the Goldfield hydrothermal district in Nevada, adjacent to the USGS Cuprite spectral benchmark.

Approach

Mineral identity is anchored to version 7 of the USGS spectral library (splib07a; Kokaly et al. 2017): every endmember is a measured library spectrum resampled to the Tanager wavelength axis, and no spectral shape is synthesized. We map the target minerals from each scene's surface-reflectance product using two complementary methods. Continuum-removed band depth (Clark and Roush 1984) measures the diagnostic absorptions directly — the 2200 nm aluminum-hydroxyl (Al-OH) doublet that separates alunite from kaolinite, the 2265 nm jarosite feature, the 2340 nm gypsum-carbonate feature, and the visible-to-near-infrared ferric-iron bands. A mixture-tuned matched filter (MTMF; Boardman 1998) then estimates per-mineral abundance against the scene's own background covariance, with an infeasibility score that suppresses spectrally implausible detections. The analytical stages run from one installed command, and the code and its shared data layer are public.

What Sentinel-2 spectral sampling blurs

The case for contiguous VSWIR is sharpest at the 2200 nm Al-OH doublet. Alunite and kaolinite (advanced-argillic versus argillic alteration, an important distinction in hydrothermal alteration mapping and ore characterization) differ in the precise position and shape of this feature. Convolving the measured library spectra with Sentinel-2's published spectral response functions reduces their separability: the alunite-kaolinite spectral angle falls from 5.1° to 2.6°, a 50% loss, because a single broad Sentinel-2 band (B12) spans the entire doublet (Figure 1). The loss is specific to the shortwave infrared, not universal; the visible-to-near-infrared jarosite-goethite contrast survives degradation, which is the control showing the effect is the doublet itself and not coarse resampling.

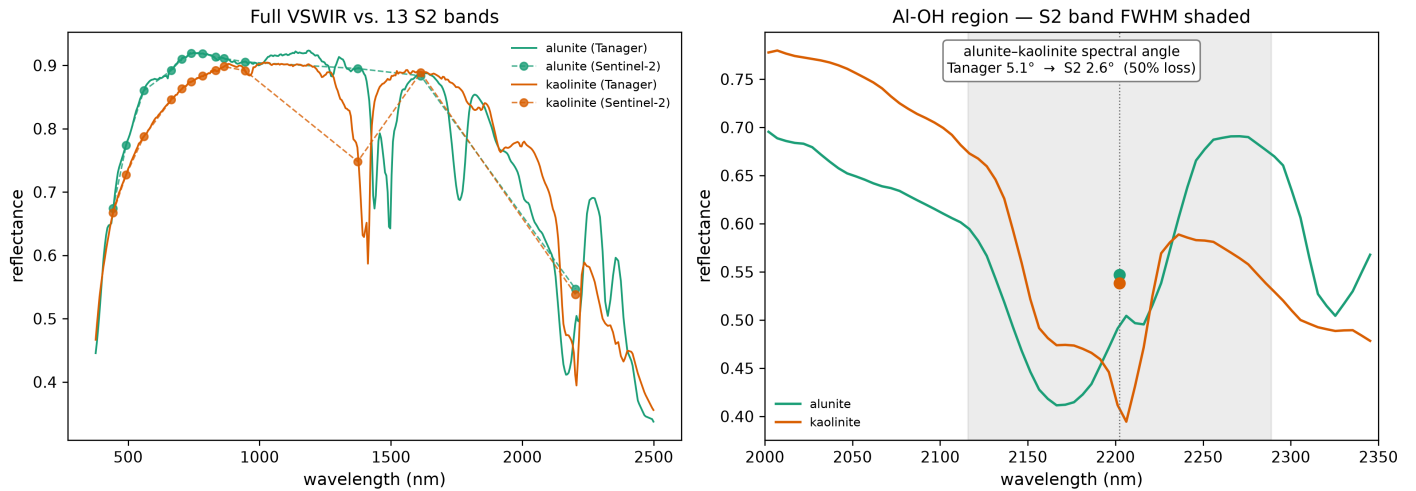


Figure 1. Tanager-to-Sentinel-2 spectral degradation at the 2200 nm Al-OH doublet.

Comparison with an independent alteration map

At Goldfield the maps were compared with the USGS ASTER alteration map of the district (Rockwell and Bonham 2017), an independent remote-sensing product with no shared calibration or acquisition. The Tanager scores separate the published alteration zones at the alteration-group level: the Al-OH band depth and the gypsum-carbonate band depth each reach a descriptive pixelwise rank-AUC of 0.78, while alunitite and muscovite MTMF each reach 0.70, with alunitite peaking in the advanced-argillic zone at Cuprite and muscovite in the surrounding sericite (Figure 2). The Fe-oxide and kaolinite layers do not separate as cleanly; the published categorical classes subdivide these phases differently than the spectral library groups them, and that disagreement is reported rather than tuned away. Bingham agrees more weakly with the regional alteration map: a porphyry system whose pervasive sericite does not partition into the published acid-sulfate zones, illustrating that agreement depends on how well the spectral classes correspond to the reference map's classes.

Cross-sensor agreement with EMIT

The same mineral-mapping pipeline was run on a pre-selected overlapping scene from NASA's EMIT imaging spectrometer, the only other spaceborne instrument with comparable coverage. Scene-mean reflectance correlates at Pearson $r = 0.962$ over 240 shared bands (spectral angle 3.72°), and all six target-mineral maps correlate positively, ranging from muscovite ($r = +0.335$) to jarosite ($r = +0.584$) (Figure 3). Because both analyses use the same code and mineral endmembers, this tests cross-sensor consistency, not independent accuracy. The map agreement is moderate, not near-unity, as expected from the acquisition-date offset and the different delivered grids. Tanager's delivered 30 m pixels cover about one-quarter the area of EMIT's ~60 m pixels; we compare the delivered products, not native instrument footprints.

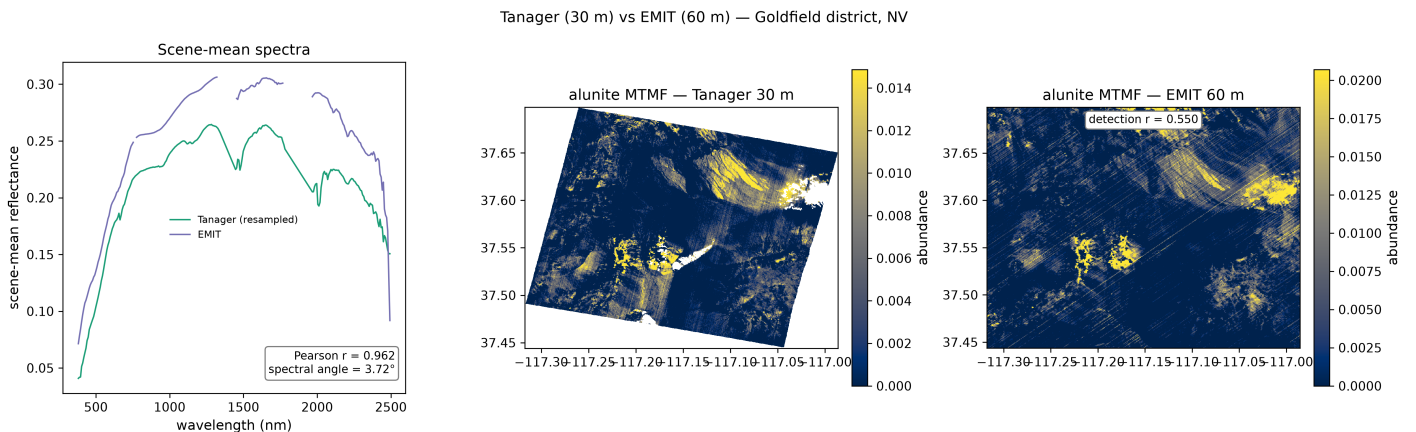


Figure 3. Tanager-EMIT cross-sensor consistency for the Goldfield scene.

Screening for acidic mine-waste conditions

Acid-generating mine waste leaves mineralogical clues at the surface, and those minerals are spectrally distinct. Jarosite forms under acidic, oxidizing, sulfate-rich conditions and provides a strong mineralogical indicator of acidic oxidative weathering; iron oxyhydroxides are generally associated with oxidation under less acidic conditions; gypsum without the acidic iron phases is more consistent with a comparatively buffered setting (Swayze et al. 2000). Each pixel is assigned an ordinal screening tier from the mineral indicator associated with the most acidic conditions detected there. The tiers rank surface mineralogical evidence for acidic conditions within the scene; they are not estimates of acid-generating capacity, pH, or acid flux. At Bingham, many of the pixels with the strongest mineralogical indicators of acidic conditions occur around the pit and tailings areas (Figure 4). The layer is a screening proxy, not a substitute for sampling.

Limits

The products have 30 m pixels, so isolated sub-pixel features and fine mine-waste structures cannot be resolved; the maps are of surface mineralogy, not bulk chemistry or depth; the spectral library can mismatch exotic phases, so the scope is held to the well-characterized alteration assemblage; and there is no field validation — the Goldfield test is against another remote-sensing product, which bounds agreement at the alteration-group level, not per-mineral abundance. The mine-waste acidity screening proxy is unvalidated at Bingham, where jarosite is absent from the regional reference map.

Impact and the case for more scenes

This is material identification against measured reference spectra — material-specific screening, not generic land-cover classification: each pixel receives a scene-relative score against a measured reference library, and the strongest supported candidate is mapped. The end users for this kind of product would be the state geological surveys and federal agencies (USGS, BLM, EPA) that characterize critical-mineral potential and mine-waste hazard; for them, a reproducible, library-anchored mineral map over a named district could serve as a candidate screening layer for field follow-up. The result also makes an archive argument: the public benefit of shortwave-infrared mineral mapping is largest over mining districts — the scenes the open archive should grow. The pipeline ships as an open tool (`tanager-minmap`, MIT) with its shared data layer at a tagged public release; the citable derivative-map archive is the one release step still open.

Figures. (1) Tanager vs Sentinel-2 band-ablation at the 2200 nm Al-OH doublet. (2) Goldfield/Cuprite alteration-group validation. (3) Tanager–EMIT cross-sensor comparison. (4) Bingham surface-mineralogical acidity screening proxy.

Data and references. Tanager STAC data (www.planet.com/data/stac), © 2024–2025 Planet Labs PBC, CC BY 4.0; NASA EMIT reflectance via LP DAAC. Boardman (1998), 7th JPL Airborne Earth Science Workshop. Clark and Roush (1984), J. Geophys. Res. 89, 6329–6340. Kokaly et al. (2017), USGS Data Series 1035. Rockwell and Bonham (2017), ASTER-derived mineral and alteration maps of the western US, USGS data release, doi:10.5066/F7CR5RK7. Swayze et al. (2000), USGS OFR 2000-0205.